

Marco Linton

PERSONAL INFORMATION

ADDRESS: Instituto de Ciencias Matemáticas, Fuencarral-El Pardo, 28049, Madrid, Spain
WEBSITE: marcolinton.github.io
EMAIL: marco.linton@icmat.es
marcolinton.math@gmail.com

ACADEMIC POSITIONS

2027- Ramon y Cajal Research Fellow, *Instituto de Ciencias Matemáticas (ICMAT)*.
2024-2027: Severo Ochoa Postdoctoral Research Associate, *Instituto de Ciencias Matemáticas (ICMAT)*.
2022-2024: Postdoctoral researcher, *University of Oxford*.
Dawid Kielak's group.

EDUCATION

2018–2022: PhD in Mathematics: *University of Warwick*.
Supervised by Saul Schleimer.
Thesis title: *One-relator hierarchies*.
2017–2018: MSc in Mathematics, *Imperial College London*.
First class honours with distinction.
2014–2017: BSc in Mathematics and Computer Science, *University of York*.
First class honours with distinction.

RESEARCH INTERESTS

Coherence, compact cores, splittings, L^2 -invariants, group cohomology, free groups, one-relator groups, hyperbolic groups, quasi-convex subgroups.

PREPRINTS

1. *Finite complete rewriting systems for graphs of free groups with applications to free-by-cyclic, one-relator, and three-manifold groups*, (with Robert Gray), 2026, [arXiv:2607.07496](https://arxiv.org/abs/2607.07496).
2. *Coherent RFRS groups* (Appendix to main article by Sam P. Fisher and Pablo Sánchez-Peralta), 2026, [arXiv:2603.16763](https://arxiv.org/abs/2603.16763).
3. *The finitely generated intersection property in fundamental groups of graphs of groups* (with Jordi Delgado, Jone Lopez de Gamiz Zearra, Mallika Roy and Pascal Weil), 2025, [arXiv:2512.12635](https://arxiv.org/abs/2512.12635).
4. *Group pairs, coherence and Farrell–Jones Conjecture for K_0* (with Andrei Jaikin-Zapirain and Pablo Sánchez-Peralta), 2025, [arXiv:2510.023518](https://arxiv.org/abs/2510.023518).
5. *The geometry of subgroups of mapping tori of free groups*, 2025, [arXiv:2510.03145](https://arxiv.org/abs/2510.03145).
6. *Pullbacks and intersections in categories of graphs of groups* (with Jordi Delgado, Jone Lopez de Gamiz Zearra, Mallika Roy and Pascal Weil), 2025, [arXiv:2508.04362](https://arxiv.org/abs/2508.04362).
7. *On the intersections of finitely generated subgroups of free groups: reduced rank to full rank*, 2021, [arxiv:2108.10814](https://arxiv.org/abs/2108.10814).

PUBLICATIONS

8. *The theory of one-relator groups: history and recent progress*, (with Carl-Fredrik Nyberg-Brodda), 2025, Chapter to appear in Recent Progress in Mathematics, KIAS Springer Ser. Math., 2., available at [arXiv:2501.18306](#).
9. *Lifting relations in right orderable groups*, 2024, Accepted for publication at J. Lond. Math. Soc., available at [arXiv:2412.17057](#).
10. *Embedding finitely generated free-by-cyclic groups in {finitely generated free}-by-cyclic groups*, 2025, *Int. Math. Res. Not.* **2026** (2026), no. 4: [MR5035094](#).
11. *Residually rationally solvable one-relator groups*. *Bull. Lond. Math. Soc.* **58** (2025), no. 1: [MR5019869](#).
12. *Group rings of three-manifold groups* (with Dawid Kielak). *Proc. Amer. Math. Soc.* **152** (2024), no. 5, 1939–1946; [MR4728464](#).
13. *On the coherence of one-relator groups and their group algebras* (with Andrei Jaikin-Zapirain). *Ann. of Math. (2)* **201** (2025), no. 3, 909–959. [MR4899802](#).
14. *Virtually free-by-cyclic groups* (with Dawid Kielak). *Geom. Funct. Anal.* **34** (2024), no. 5, 1580–1608; [MR4792841](#).
15. *Hyperbolic one-relator groups*. *Canad. J. Math.* **78** (2026), no. 1, 35–61. [MR5020065](#).
16. *One-relator hierarchies*. *Duke Math. J.* **174** (2025), no. 4, 747–802. [MR4905535](#).
17. *The fully compressed subgroup membership problem*. *J. Algebra* **628** (2023), 562–583; [MR4579901](#).

INVITED CONFERENCE TALKS

- | | |
|-----------|--|
| Jul. 2026 | Arithmetic Groups, High-dimensional Expanders and L^2 -invariants (Brin Mathematics Research Center): <i>The Hanna Neumann conjecture and submultiplicativity of L^2-Betti numbers.</i> |
| Jun. 2026 | Young Geometric Group Theory XIV (KIAS): <i>Hanna Neumann bounds for cubulated locally quasi-convex hyperbolic groups.</i> |
| Jun. 2026 | Sapir distinguished Lecture at GAGTA (CUNY): <i>One-relator groups: past, present and future.</i> |
| Aug. 2025 | Algebraic and Analytic Methods in Group Theory in Warsaw (IMPAN): <i>Coherence of group pairs.</i> |
| Aug. 2025 | Operators, Graphs, Groups Colloquium (Newton Institute): <i>The geometry of subgroups of free-by-cyclic groups.</i> |
| Jul. 2025 | JAE School of Mathematics (ICMAT): <i>How complicated can a group with a single defining relation really be?</i> |
| Jun. 2025 | Geometric Group Theory and Related Topics (Peking University): <i>Quasi-convex subgroups of free-by-cyclic groups.</i> |
| Jan. 2025 | RSME's 7th Congress of Young Researchers (University of the Basque country): <i>The geometry of free-by-cyclic groups.</i> |
| Sep. 2024 | Combinatorial Non-positive Curvature (Banff International Research Station): <i>Lifting relations in right orderable groups.</i> |
| May. 2024 | Barcelona Weekend in Group Theory (Universitat Politècnica de Catalunya): <i>The relation gap and relation lifting problems for groups with cyclic relation modules.</i> |

- Apr. 2024 Groups, Topology and Applications in Gran Bilbao V (University of the Basque Country): *The relation gap and relation lifting problems for groups with cyclic relation modules.*
- Feb 2024 GAGTA (CIRM): *The coherence of one-relator groups.*
- Jun. 2023 Groups and Computation (Stevens Institute of Technology): *How to prove one-relator groups are coherent.*
- May 2023 Geometry of subgroups (Centre de Recherches Mathématiques): *The coherence of one-relator groups.*
- May 2023 One-day special NYGT event in memory of Gilbert Baumslag and Benjamin Fine (City University of New York): *One-relator groups, virtually free-by-cyclic groups and coherence.*
- Oct. 2022 Groups and Geometry in the South east (University College London): *Hyperbolic one-relator groups.*
- Sep. 2022 Modern Advances in Geometric Group Theory (University of Manchester): *Simplifying Gersten's conjecture.*
- Aug. 2021 World of groupcraft (online conference): *Fibre products, free groups and regular languages.*

SEMINAR TALKS

- May. 2026 Long-distance seminar on Geometric Group Theory in Mexico (Centro de Ciencias Matemáticas): *TBA.*
- Mar. 2026 McGill Geometric Group Theory (McGill University): *Characterising local quasi-convexity.*
- Mar. 2026 Geometry and Topology Seminar (University of Bristol): *Characterising local quasi-convexity.*
- Jan. 2026 Algebra, Geometry and Topology Department Colloquium (Universidad Complutense de Madrid): *The geometry of one-relator groups.*
- Nov. 2025 Geometry Seminar (Max Planck Institute Leipzig): *The geometry of mapping tori of graphs.*
- Nov. 2025 Glasgow Geometry & Topology seminar (University of Glasgow): *Coherence of groups pairs.*
- Apr. 2025 Geometric Group Theory Seminar (University of Bonn): *The geometry of free-by-cyclic groups.*
- Apr. 2025 Madrid Group Theory Seminar (ICMAT): *The geometry of free-by-cyclic groups.*
- Dec. 2024 Wrocław Geometry Seminar (University of Wrocław): *The geometry of free-by-cyclic groups.*
- Nov. 2024 KIAS-Springer Lectures (Korean Institute for Advanced Study): *An overview of recent progress in the theory of one-relator groups.*
- Oct. 2024 BG-UToledo Geometry and Topology Seminar (Bowling green University and University of Toledo): *The relation gap and relation lifting problems.*
- Jul. 2024 Junior Topology and Group Theory Seminar (University of Oxford): *The relation gap and relation lifting problems.*
- Mar. 2024 Géométrie dynamique (Université de Lille): *The coherence of one-relator groups.*
- Mar. 2024 Geometry and Topology Hybrid (University of Warwick): *The coherence of one-relator groups.*
- Mar. 2024 Algebraic Topology Seminar (University of Warwick): *The relation gap conjecture for groups with cyclic relation modules.*
- Feb. 2024 Topology Seminar (Princeton University): *The coherence of one-relator groups.*

- Jan. 2024 Geometric Group Theory Seminar (Univeristy of Bonn): *The coherence of one-relator groups.*
- Nov. 2023 Geometry and Topology Seminar (University of Bristol): *The coherence of one-relator groups.*
- Nov. 2023 Group Theory Seminar (Instituto Ciencias Matemáticas): *Virtually free-by-cyclic groups, one-relator groups and coherence.*
- Nov. 2023 Group Theory Seminar (Universitat Politecnica de Catalunya): *The coherence of one-relator groups.*
- Oct. 2023 Purdue Geometry and Geometric Analysis Seminar (Purdue University): *Coherence, non-positive immersions and L^2 -Betti numbers.*
- Oct. 2023 Manchester Algebra Seminar (University of Manchester): *Coherence, non-positive immersions and L^2 -Betti numbers.*
- Mar. 2023 Geometry and Analysis on Groups Research Seminar (Universität Wien): *Virtually free-by-cyclic groups.*
- Dec. 2022 Algebra, Geometry and Topology Seminar (Heriot-Watt University): *Hyperbolic one-relator groups.*
- Nov. 2022 Geometric Group Theory Seminar (University of Oxford): *Hyperbolic one-relator groups.*
- Nov. 2022 Fudan Topology Seminar (Fudan University): *Hyperbolic one-relator groups.*
- Oct. 2022 Dagger seminar (University of Warwick): *Posion subgroups for hyperbolic groups.*
- May 2022 Pure Postgraduate Seminar (University of Southampton): *Poison subgroups. for hyperbolic groups.*
- Apr. 2022 Geometry and topology online (University of Warwick): *Hyperbolicity of certain one-relator groups.*
- Apr. 2022 Algebra seminar (University of Bristol): *Hyperbolicity of certain one-relator groups.*
- Apr. 2022 New York Group Theory Seminar (City University of New York): *Hyperbolicity of certain one-relator groups.*
- Apr. 2022 Geometry and Analysis on Groups Research Seminar (Universität Wien): *Primitivity rank, one-relator groups and hyperbolicity.*
- Mar. 2022 Group theory seminiar (Universitat Politecnica de Catalunya): *Primitivity rank, one-relator groups and hyperbolicity*
- Nov. 2021 Geometric group theory seminar (University of Cambridge): *When does a one-relator group have a quasiconvex hierarchy?*
- Oct. 2021 Dagger seminar (University of Warwick): *Dynamics and isoperimetric inequalities for the double mapping torus.*
- Apr. 2021 Pure PGR seminar (University of Southampton): *An ode to the fibre product graph.*
- Jan. 2020 Bridges seminar (University of Bristol): *Hierarchies for one-relator groups.*
- Dec. 2019 Junior geometry and topology seminar (University of Oxford): *Hierarchies for one-relator groups.*
- Dec. 2019 Postgraduate seminar (University of Warwick): *Hierarchies for one-relator groups.*
- Nov. 2019 Dagger seminar (University of Warwick): *Hierarchies for one-relator groups.*
- Feb. 2019 Gateway seminar (University of Warwick): *Dehn filling in relatively hyperbolic groups with torsion.*

CONTRIBUTED TALKS

- Feb. 2023 Young Geometric Group Theory (University of Münster): *An overview of one-relator groups*. (Poster).
- Jul. 2019 Young Geometric Group Theory (University of the Basque country): *The word problem in graphs of free groups*.

MINICOURSES

- Sep. 2027 I will give a minicourse at the Ventotene International Workshop ‘Negatively Curved Immersions, Coherence and the Surface Subgroup Conjecture’.
- Oct. 2026 I will give an 8 hour minicourse at the Shanghai Center for Mathematical Sciences on *Free-by-cyclic groups and their subgroups*.
- Jun. 2026 I gave a 3 hour minicourse at ‘Advanced school: Aspherical 2-complexes and their fundamental groups: past, present, and future’ at the ICMAT on: *2-complexes with vanishing L^2 -Betti numbers*.
- Nov. 2024 I gave a 3 hour minicourse at the ‘KIAS workshop on One Relator groups and other Aspects of GGT (KOREA-GGT)’ on: *One-relator groups and the coherence property*.
- Jul. 2024 I gave a 9 hour graduate course at the ‘Topological and analytic methods in group theory’ conference at the University of Hawaii on: *Coherence, homological coherence and one-relator groups*.
- Apr. 2024 I gave an 8 hour minicourse to experts at the conference ‘Groups, Topology and Applications in Gran Bilbao V’ at the University of the Basque country on: *The coherence of one-relator groups*.

EXTENDED RESEARCH VISITS

- 28 Jul. – 8 Sep. 2025 Newton Institute, Cambridge, UK. Operators, Graphs, Groups thematic programme. Invited participant. Colloquium speaker.
- 15 May – 30 Jun. 2023 Centre de Recherches Mathématiques, Montreal, Canada. Geometric Group Theory thematic programme. Conference speaker.

SHORT RESEARCH VISITS

- 22 – 27 Mar. 2026 McGill University, Canada. Visiting Jean-Pierre Mutanguha for collaboration and seminar talk.
- 8 – 13 Mar. 2026 University of Bristol, UK. Visiting Joseph MacManus for collaboration and seminar talk.
- 13 – 19 Apr. 2025 University of Bonn, Bonn, Germany. Visiting Sam Hughes for collaboration and seminar talk.
- 1 – 7 Dec. 2024 Wroclaw University, Wroclaw, Poland. Visiting Motiejus Valiunas for collaboration and seminar talk.
- 14 – 27 Apr. 2024 University of the Basque country, Bilbao, Spain. Visiting Montse Casals-Ruiz for collaboration and giving a mini-course at the conference.
- 10 – 16 Mar. 2024 University of East Anglia, Norwich, UK. Visit to Robert Gray for collaboration. Funded by my LMS Research in Pairs grant.
- 11 – 17 Feb. 2024 Princeton University, Princeton, USA. Visiting Jean-Pierre Mutanguha for collaboration and seminar talk.

- 7 – 13 Jan. 2024 University of Bonn, Bonn, Germany. Visiting Giles Gardam for collaboration and seminar talk.
- 12 – 18 Nov. 2023 ICMAT, Madrid, Spain. Visiting Andrei Jaikin-Zapirain for collaboration and seminar talk.
- 5 – 11 Nov. 2023 Universitat Politecnica de Catalunya, Barcelona, Spain. Visiting Jordi Delgado for collaboration together with Jone Lopez de Gamiz, Mallika Roy and Pascal Weil.
- 12 – 18 Mar. 2023 Universitat Wien, Vienna, Austria. Visiting Christopher Cashen for collaboration and seminar talk.
- 6 – 12 Mar. 2022 Universitat Politecnica de Catalunya, Barcelona, Spain. Visiting Enric Ventura for collaboration and seminar talk.

SUPERVISION

- Alba Gonzalez Gonzalez (Masters student at Universidad Autónoma de Madrid), now pursuing a PhD at Oxford.
- Miguel Angel Royo Miguel (JAE summer school research student from Universidad de Zaragoza).

PANELS AND COMMITTEES

- Director of the thesis committee for Pablo Sánchez Peralta's PhD defence (Universidad Autónoma de Madrid).
- Scientific committee for the conference 'Advanced school: Aspherical 2-complexes and their fundamental groups: past, present, and future.'
- Academic career panel at conference 'Topological and analytic methods in group theory.'
- Committee for Sam Fisher's confirmation of status (University of Oxford).

TEACHING

In 2025 I taught the advanced algebra course at the Universidad Autonoma de Madrid (UAM) on *Geometric Group Theory*.

SUPPORT CLASSES

Between 2018 and 2022 I worked as a teaching assistant for the University of Warwick. I provided support classes and marked assignments for the following courses:

- Discrete Mathematics and its Applications I.
- Algebra II.
- Geometry.
- Introduction to Topology.
- Algebraic Topology.

AWARDS AND GRANTS

- Awarded a Ramon y Cajal grant (2025). Ranked number 1 in mathematical sciences out of 93 applicants, with a score of 99/100.
- Awarded the Sapir Distinguished Lecture (2025).
- LMS research in pairs grant to visit Dr. Robert Gray (2023).
- Departmental funding for supervision of undergraduate summer project, *Finding solutions to the Post correspondence problem for free groups*, University of Oxford (2023).
- Winner of the prize for best MSc pure mathematics student, Imperial College London (2018).
- Winner of the Kathleen Ryan project prize for the best overall final year BSc project performance, University of York (2017).

ORGANISATIONAL ACTIVITIES

- Organiser for the conference *Young Geometric Group Theory XV* at the ICMAT in 2027.
- Organiser for the conference *Groups in Madrid* at the ICMAT in 2025.
- Organiser for the Dynamics and Group Geometry Early Researchers (DAGGER) seminar at the University of Warwick.
- Organiser for a reading group on 3-manifolds, following *An introduction to geometric topology* by Bruno Martelli.
- Organiser for a reading group on CAT(0) cube complexes, following *The structure of groups with a quasi-convex hierarchy* by Dani Wise.
- Organiser for a reading group on JSJ decompositions, following *JSJ decompositions of groups* by Vincent Guirardel and Gilbert Levitt.

REFEREEING

I have refereed for Michigan Mathematical Journal, Journal of the London Mathematical Society, Proceedings of the Royal Society of Edinburgh, Journal of Algebra, Israel Journal of Mathematics, Journal of the European Mathematical Society, Mathematische Zeitschrift and International Journal of Algebra and Computation.

I have also given quick opinions for the Journal of the European Mathematical Society, for Geometry and Topology, for the Journal of the London Mathematical Society and for Groups, Geometry and Dynamics.

I am also a reviewer for Mathscinet.

LANGUAGES

- English: mothertongue
- Italian: fluent.
- Spanish: working knowledge.